

Comprehensive Evaluation of Job satisfaction using Deep Learning Approaches

Devanshu Singh

Abstract

With the ongoing development of number of companies that are growing in market these days, Employees are looking for companies that are having highest number of employees satisfaction. Together with these many existing employees or past employees post reviews on famous platform such as Glassdoor Considering the need an accurate deep learning-based approach that can help job seeker to get a true review about the organization they are planning to join. The research findings of this paper evaluate the capability of different Deep learning models in judging the job satisfaction on the basis of textual reviews posted on Glassdoor Job dataset. By combining input of reviews, pros and cons together with ratings provided we compare Distil BERT and LSTM model to evaluate overall satisfaction score of employees on the scale of 1 to 5. The main objective of the study is to analyse how accurately Deep learning models can judge employee sentiments. Also, this study focuses on determination of linguistic pattern and trends related to Job satisfaction and dissatisfaction by including SHAP with Distil BERT which determine key linguistic contributors to calculated score. The performance of the advanced model (Distil-BERT+SHAP) is compared with state-of-the-art deep learning models such as Bi-LSTM and the results demonstrate that the Distil-BERT model performs better.

Keywords – SHAP, Distil BERT, Bi-LSTM, Deep Learning

I. Introduction

As the number of companies are growing, satisfaction of employees is one such parameter that is of utmost importance for growth of the organization specially for retaining skilful employees. Together with this employee are also

focusing not only on salary package and big company names they also look forward to reviews posted on the common platforms like Glassdoor which provide them broad insight about work culture of said company. These common platform holds a wide variety of reviews about the company such as Reviews, rating which might have pros and cons section as well and these sections showcase the sentiments of employees after working in that particular organization. However, it might be a time-consuming task to manually judge large number of reviews and generate insights.

With this problem there is need to provide technologically advanced solutions, as this for judgement of sentiments and human thoughts based on textual reviews Natural Language processing and Deep learning-based approaches are ideal choice. The primary motive of this search is not only analysing sentiment but rating them on standardized scale of 1-5 which will be helpful for the organization job candidates and HR departments to make better decisions. Also, this report will follow a CRISP-DM based architecture to follow a structured approach to answer the business problem with core use technological aspects. The detailed steps will be explained in upcoming sections.

This study focuses on answering Research question “How accurately can we predict employee job satisfaction from textual reviews using deep learning models, and what linguistic patterns and temporal trends correlate most strongly with high or low satisfaction? To cater this situation performance of two major deep learning models DistilBERT which is computation wise lighter version of actual BERT which is known for better efficiency and Bi-LSTM

model which has features that solves textual problem. Both the model is trained with a large dataset of Glassdoor reviews about companies and correlate result on the basis ratings provided by the employees.

Also, to provide Explainability of models DistilBERT results was fed into SHAP (Shapley Additive Explanations), It focuses on judging the intensity & contributions of each word for the predictions models result. It basically offers a clear picture of the deep Learning model used. Also, Distil BERT with SHAP is one such approach supports higher flexibility, accurate and explainable predictions which makes it ideal to handle variations in textual inputs without need of much alterations.

This research is beneficial for both Technological and Applications aspects. Application wise its helpful for employee to choose good organization on the basis of detailed analysis of reviews and ratings. And, technologically it compares advanced Deep learning-based models DistilBERT and Bi-LSTM which offers accurate predictions. On the whole, this research provides better solutions for analysing employee sentiments through textual data in terms of scalability, accuracy and explainability.

For running Distil BERT, we have use python version 3.10.0 & tensor flow 2.11.0 and NumPy version less than 2.0 for compatibility with our suggested model.

II. Literature Review

The use of Deep learning models for analysing reviews or text has gained popularity and topic of the research for many in recent times, driven by the necessity for fast and accurate prediction of job reviews or general reviews to make well informed decisions. Specially for the Job market employees wants to look for the job satisfaction score of the organization to choose a better option. This section critically reviews cutting edge research that tackles important issues in domain of Job Market and usage of Distil BERT and Bi-LSTM for textual analysis.

Sentiment classification for employees' reviews has been explored by [1] that suggested hybrid approach for classification model which is RV-SGDC that uses TF-IDF vectorization and a Stochastic Gradient Descent-based classifier for analysing sentiment polarity in employees' reviews. However, their research is only directed towards binary classification and majorly misses accurate prediction and interpretability. On this basis our

research a step forward, that utilizes advance Deep learning models for predicting sentiment on the scale of 1-5 incorporating SHAP to providing a transparent model.

Going forward, [2] also conducted a detailed comparison analysis of deep learning algorithms in sentiment analysis, It particularly highlights Bi-LSTM capability in judgement of textual analysis. This works as a basic model for our study that focusses on analysing employees' reviews. Also, [3] uses SVM on Glassdoor reviews for judging accurate sentiment classification, it mainly uses Random Forest algorithm. Their work might be effective but fails to explore deep learning models with explainability of model, where our approach is using deep learning techniques and SHAP based interpretation.

As the need and interest for utilizing employee review from common platforms like Glassdoor increases to generate insights Many are using such data one of them is [4] that initiated research in this domain their study mainly uses combination of both deep learning and machine learning based algorithms and comes on a conclusion that LSTM based approaches is far better than Logistic regression and traditional random forest model. Our research is building on this foundation that mainly includes training advanced model such as DistilBERT and using SHAP for enhancing interpretation and explainability of model together with improved transparency.

As the need of sentiment analysis grows getting idea about employee experiences, [5] uses LSTM based approach for classification of sentiment from given feedback provided by employees. Also, considering the need of such system in expanding and understanding Human resource analytics by [6] works on a hybrid approach that combines both machine Learning and Deep learning methods for judgement of employee sentiments across different area. Their research was mainly directed towards understanding emotional tone for better output. In combination both of these shows use of hybrid approach effectiveness in sentiment prediction which is base of our research.

Considering the growth and advancement in the domain of analysing sentiments many have incorporated use of Hybrid approaches for enhancement of accuracy in classification. Also, [7] proposed an approach where they are using combination of pre-trained BERT, Bi-LSTM and Bi-

GRU algorithms, which showcased better performance across distinct datasets. On the similar ground [8] also provided hybrid method that integrates BERT and DistilBERT for text Bi-LSTM and CNN for classification purpose. This method shows superior performance in sentiment classification as they were able to capture both contextual and semantic features of text. Building upon this our study compares both Distil BERT and Bi-LSTM models to predict employee satisfaction scores and incorporates SHAP with Distill Bert for effective explainability of model.

Based on analysis with comparison to Machine learning based approaches deep learning models performed better in terms of capturing contextual meaning for sentiment related tasks. As per study initiated by Kusal et al. [9] a comparison of traditional Deep learning-based model and advanced transformer-based model such as BERT is conducted and the result were quite promising where Transformer based model performed extremely well on contextual and accuracy parameters. Together with this according to another relatable research Bhuvaneshwari et al. [10] proposed Hybrid model which is combination of Bi-LSTM and self-attention-based CNN, which actually performed well. Both of this research specially comes out with a fact that both Bi-LSTM transformer-based models are ideal choices. On the basis of motivation derived from these we are try to compare Distil BERT and Bi-LSTM for understanding Employee sentiments and further increasing its interpretability employing SHAP with DistilBERT for model explainability

As the user generated content is getting exploded in market in almost everywhere on internet, several techniques such as transfer -learning are being used for detection of sentiments from textual data. To support this [11] performed detailed evaluation for detecting reviews and for this purpose transformer-based models such as BERT, RoBERTa, ALBERT, and DistilBERT are used. And their prediction shows that all of the models performed well and DistilBERT shows good balance of both accuracy and computation. This marks the importance of light-weight models in NLP applications where speed and accuracy are important. Taking Inspiration from these models our study uses DistilBERT for processing reviews fetched from Glassdoor datasets that aims to produce high accuracy and ensuring explainability by using SHAP with it.

With the rise of social media usage various common platform for judgement of companies is becoming important for behavioural analysis. In this regards, [12] has actually used Linked data for analysing sentiments of employees using NLP approach, that shows the importance of user-generated textual data. Their research is actually baselining foundation of more advanced applications of sentiment analysis. Also, another research done by [13] introduced a Bi-LSTM framework which is designed to handle imbalances in the datasets from reviews. Their study shows that accuracy is being improved in detection of niche sentiments while addressing noise in textual data. These studies actually tell the importance of deep learning models explored by us which is comparison of Bi-LSTM and advanced Distil-BERT and introducing incorporating SHAP with it to provide explainability of model

As the research in similar domain evolves, there is focus on solutions that combines deep learning traditional models this actually benefits quality of prediction and domain adaption. To support this [14] describes an approach that uses Support Vector Machines (SVM) and Structural Topic Modelling (STM) for getting key factors of job satisfaction. This actually supports predictive and topic level analytics. On the similar basis [15] provided a model that uses Bi-LSTM and GRU to further improve performance on reviews types of datasets. Both of these models work on long range dependencies and imbalances in class which is issue of real datasets. Taking inspiration from these our work compares Distil BERT and Bi-LSTM for sentiment prediction wherein SHAP provides an added advantage that works on explainability of model.

Also, Traditional Job surveys focus on fixed type of questions which impacts the quality of generated insights. To cater [16] showcase why loosely open responses produced more accurate results and the tools that they used are Sentiment and LIWC. This level of growth of research in prediction of intensity of sentiments in prediction of job satisfaction together with this use of advanced transformer-based models such as Distil BERT and Bi-LSTM shows an evolution of research towards more interpretable context sensitive methods. By combining Distil BERT with SHAP we are aiming to provide a system or framework that not only provide sentiment analysis but it also explains it and provide intensity of sentiments.

III. Methodology

In this section we will explain, methodology of our approach for prediction job satisfaction of employees on the basis of Glassdoor data. Our work is structured mainly under CRSIP-DM (Cross-Industry Standard Process for Data Mining) it provides clear roadmap for data-driven projects and focus on six main stages Business understanding, Data Understanding, Data preparation, Modelling, Evaluation and Deployment.

A. Business Understanding:

The main focus of our research work is the development of smart and self-explainable system that works on automatic assessment of job satisfaction of employee by using textual reviews from platform like Glassdoor. This research is actually generated to help employees and HR to make well informed decisions. Our system is unique in way that it doesn't only relay on rating, it focusses on generating insights using textual reviews, pros, cons and rating by using advanced deep learning approaches with explainable AI(XAI) methods. To improve the transparency of results the selected model incorporates SHAP that makes system interpretable for real world use.

B. Data Understanding and Exploration

For generating initial better understanding of our dataset, According to Fig.1 we have created word cloud for the pros and cons section to visualize the words are that are arriving more frequently in datasets. Where, few words like flexibility and teams were more frequently used while some common words such as workload appears in cons, this helps us to see sentiments in the first analysis.



Fig 1: Word cloud

According to Fig.2 A heatmap is used for judging relationship among numerical ratings features. It depicts that overall ratings are correlated to culture and values which shows key factor of employee satisfaction

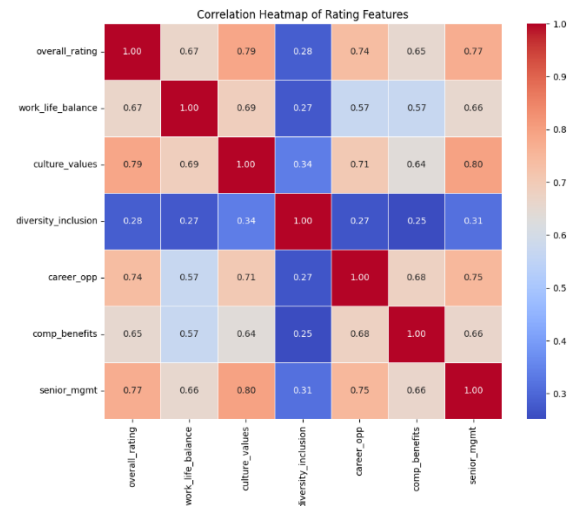


Fig.2 Correlation of Rating Features

As per Fig. 3 we are trying to understand based on initial data how satisfaction will vary across different roles for most important job titles. This helps to get an idea or details about roles specific strength and weakness within organization.

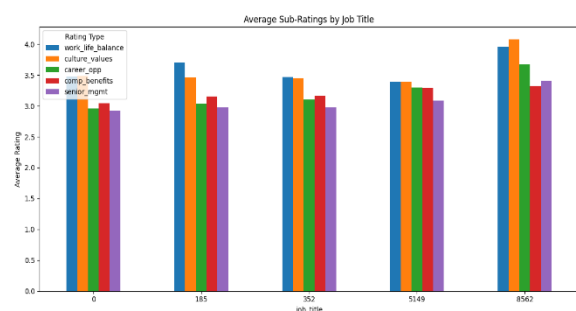


Fig.3 Average sub-ratings by Role

As per Fig.4, given box plot is used to compare sub rating among varied employees outlook. This actually depicts that person which has positive outlook usually provides high rating as comparison to one with opposite trait.

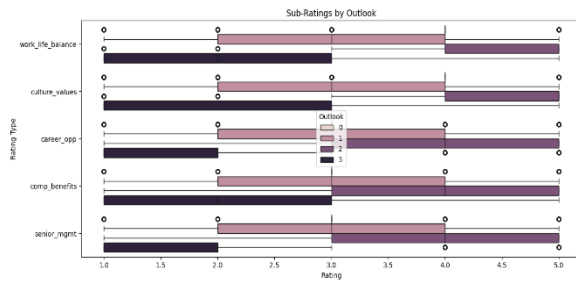


Fig.4 Sub-ratings by Outlook

C. Data Pre-processing

Starting with pre-processing steps missing values in data are visualized using bar-plot as shown in Fig.5, This will be helpful in highlighting features with missing data.

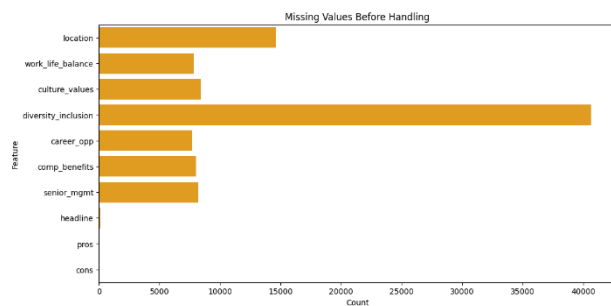


Fig.5 Missing value

For preparing model for modelling, many steps are applied to pre-process data before putting that into model. To begin with key field that contains text-based data were cleaned by removing noise and Punctuation. The Text was then tokenized and filtered for extracting meaningful words. Also, together with this sentiment scores were predicted. Going forward, missing values are replaced using median or default values.

Together with this numeric feature were standardized for normalizing the range, improving performance of the model. Fields like recommend, CEO approval, outlook, and current employment status are labelled as encoded. This step was actually required to use them in deep learning-based models.

D. Model Explanation

In this stage we will explain various stage of Model that we have applied for both Bi-LSTM and Distil BERT with SHAP

1. Bi-LSTM

For generating the effective contextual meaning of employee reviews, a B Bidirectional Long Short-Term Memory (Bi-LSTM) was applied. It is an RNN based advanced model which is having capability of

learning in both forward and backward direction, which is ideal for understanding patterns in text.

Step 1: Data preparation: For this model text inputs were created with a combination of headline, pros & cons along with this target variable overall rating was adjusted so that it starts from 0. Also, categorical column such as Job, title, location was encoded and labelled and their structured ratings were grouped into another numerical feature column.

Step 2: Tokenization and Splitting

As a next step, text data was split into token and then a padding of uniform length is created that uses 95% of sequence present. Also, the given dataset is further divided into training testing tests, which has various parallel branches text, jobs, location and rating

Step 3: Model Architecture

A multi-input Bi-LSTM model was created wherein Text input was passed through embedding layer, Job and location was passed through embedding and LSTM layer. And, structured features were combined with all related outputs which is further transferred into dense most layer of model with dropout. The final output in this step uses softmax approach for prediction

Step 4: Hyperparameter Tuning

For applying optimization step to our model, a manual grid-based search was performed on the LSTM units, learning rate, dense layer size. For this step best configuration is selected which will be based on the validation accuracy.

Step 5: Final Training

For effective modelling the best model was as a part of final training best model was selected with early stopping and checkpointing method to achieve optimal performance.

2. Distil BERT with SHAP

It is an advanced deep learning model which is lighter but effective version of BERT. We are applying this to our textual data analysis. For this we have designed an automatic pipeline

Step 1: Input Preparation and Feature Engineering

As first step text based reviews will be sent into processor using DistillBERT tokenizer which have truncation and padding which is dynamic in nature. Each review was firstly encoded into input_ids 7

attention masks. Along with this total 14 features which are structure was scaled. Also, an important target variable was labelled shifted from 0 to 4. At the end all features were split into training and test

Step 2: Model Architecture

In this step a customised keras model, BERT fusion was designed to work with structured input features of DistilBERT. Also, in parallel manner structured input will be sent to dense layer of model and projected in same space. These were then concatenated, normalized, passed through dense layers with dropout, and finally classified into 5 categories via a SoftMax output layer.

Step 3: Hyperparameter Tuning via Manual Grid Search

For optimizing the performance of model, a manual grid-based search is conducted over Units in dense layer (64,128), dropout rates (0.2,0.3), optimizers (Adam) and Batch sizes (16,32). In this scenario the model was first trained and then evaluated for accuracy. The best performing configuration was selected at the end.

Step 4: Final Model Training

In this step whenever best parameter is selected, the final model was trained on complete training set. In this case computed class weights were applied for handling class Imbalance. Together with this early stopping was used to stop training when no performance improvement is seen this actually saves from overfitting.

Step 5: Fine-Tuning DistilBERT

In starting model layer were frozen to enhance the training step. At this step layers were unfrozen and fine-tuned to improve learning. This step is important as because of this model adapt DistilBERT parameters which enhances classification accuracy.

Step 6: Model Interpretation Using SHAP

Also, for ensuring transparency of Model SHAP was applied for explainability. This method was used for explanation of contribution of text and structured features. This step actually tells how a specific word contributes to model satisfaction score which makes model explainable.

E. Model Evaluation

1. Bi-LSTM:

As shown in Fig.6 the model shows test accuracy of 63.97% that indicates moderately strong performance in the prediction of satisfaction level of employees. The macro F1 score of 0.62 suggests consistent performance across various classes. These results show the model efficiency for capturing clear sentiment and distinguishing confusing words.

Classification Report:				
	precision	recall	f1-score	support
0	0.81	0.67	0.73	1318
1	0.44	0.53	0.48	942
2	0.50	0.64	0.56	1710
3	0.55	0.59	0.57	2593
4	0.85	0.70	0.77	3476
accuracy			0.64	10039
macro avg	0.63	0.62	0.62	10039
weighted avg	0.67	0.64	0.65	10039

Rounded Accuracy: 0.6397

Fig. 6: Bi-LSTM Classification Report

Along with this, Fig. 7 represents a confusion matrix that shows how well model is capable to classify each satisfaction level. Wherein, the matrix compares actual label with predicted label. This heatmap produces clear picture of which satisfaction model is able to predict successfully and where it is struggling.

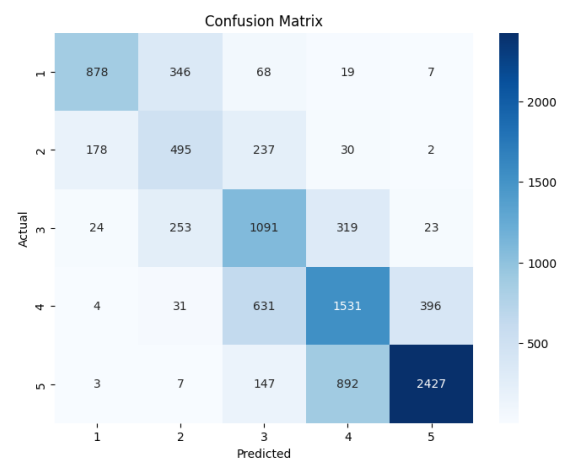


Fig.7 Confusion Matrix Bi-LSTM

Also, this confusion matrix is able to predict satisfaction at very extreme stage like very dissatisfied and very satisfied and 827, 2427 correction predictions However, it shows confusion in middle classes which is not required. For example, many classes 3 words were understood as class 4 or 2. This shows that model efficiency able to capture clear words however, struggle with subtle differences in moderate satisfaction level.

2. Distil BERT

a. Model based Evaluation

As per Fig. 8 Distil BERT class-based performance indicates that model perform best on certain classes like 0, 4, 5. It has F1 score of 0.72 and 0.74. These high scores suggests that model is good in identifying sentiments on the scale of 0 to 5.

The **macro average F1-score of 0.45** and **weighted average F1-score of 0.51** reflect this imbalance. While DistilBERT captured strong sentiment patterns well, its ability to differentiate between **subtle or moderate sentiments** (especially in classes 1 and 3) was limited. This indicates the need for further model tuning or perhaps additional training data to better learn mid-scale satisfaction level.

Classification Report:				
	precision	recall	f1-score	support
0	0.69	0.77	0.72	1318
1	0.50	0.00	0.00	942
2	0.42	0.66	0.51	1710
3	0.49	0.18	0.26	2593
4	0.63	0.90	0.74	3476
accuracy			0.57	10039
macro avg	0.54	0.50	0.45	10039
weighted avg	0.55	0.57	0.51	10039

Fig. 8. Classification Report

Also, we have evaluated model predictions by rounding both calculated and true label to nearest whole number in the range of 1 to 5. It then computes accuracy which is 60.54 for Distil BERT model. The rounded accuracy actually shows clear idea how often model fall into correct rating and in real world applications its needed.

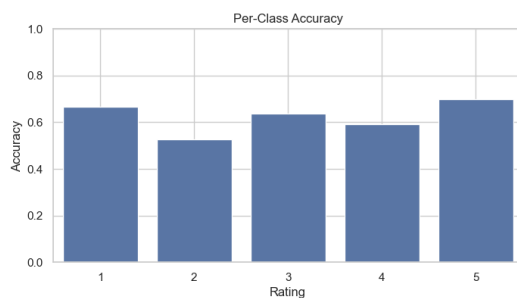


Fig.9 Per class Accuracy of Distil BERT

On the basis of Fig. 9 per class accuracy of Distil BERT model is showing consistent performance across all levels compared to Bi-LSTM. Also, Distil BERT shows to achieved higher accuracy on majorly extreme classes (5) and class 1. Also, it maintains moderate performance across middle classes as well. Contrary Bi-LSTM struggled to performed on middle classes This strongly Indicates its capability to capture subtle linguistic hints across

all levels and a reliable solution over LSTM approach.

b. SHAP – For explainability of Model

To accurately address sensitive application area of Distil BERT where transparency of model is desired, we have incorporated SHAP based approach in detail which integrates interpretation capability to model. SHAP contributed to assign review to each word like how much it was responsible for final predictions. This will be helpful for HR Teams for prediction of which part of employee's feedback are responsible for satisfaction score or for employees to judge model transparency and reliability by combination of SHAP with Distil BERT model is now accurate, transparent, explainable and reliable.

Also, as per Fig.10 Red shows positive words blue shows intensity of negative words which explains model transparency and reliability.

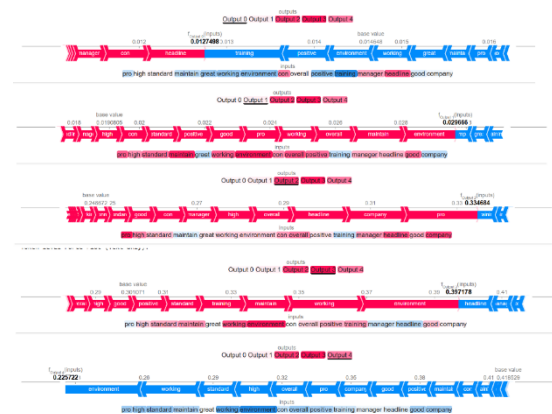


Fig. 10: SHAP output

As Per Fig.11 the bar graph shows the top most contributing word for the model to offers transparency that shows which words matter most which matters least

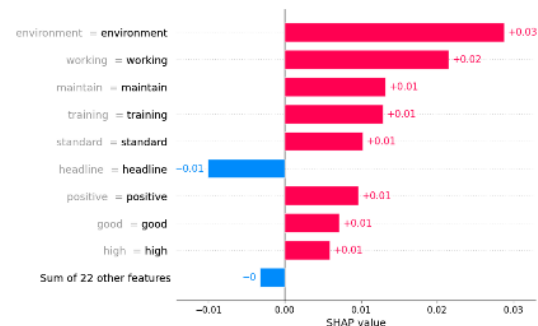


Fig.11 SHAP Summary

c. Overall Application wise Insights:

These insights showcase application on data generated after applying Distil BERT model on employee review dataset.

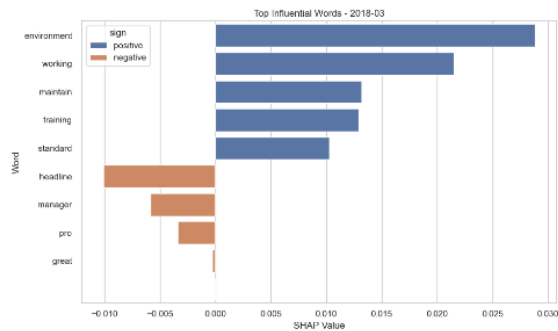


Fig: 12 Top influential words by SHAP

As per Fig .13, after 2020 there was major improvement in job satisfaction this might be due to the work from setting during covid era it might be one of the influential factors.

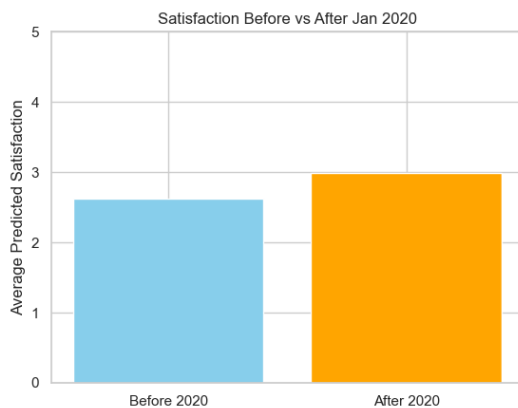


Fig.13: Satisfaction score Before and after 2020

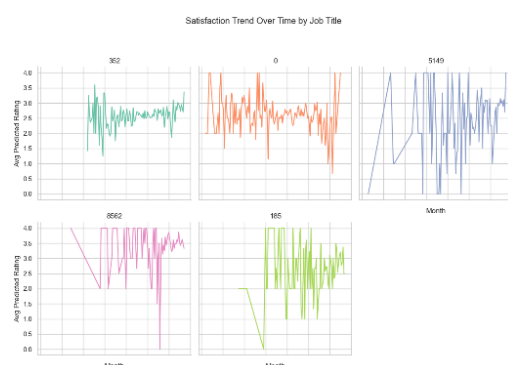


Fig.14. Satisfaction Trends over the Years by Job titles

IV. Conclusion and Future Work

In this research project we have compare two complex model Bi-LSTM and Distill BERT with SHAP for predication satisfaction score based on the reviews on the Glassdoor platform. While both

model shows strength Distil BERT performs better than Bi-LSTM in terms of handling extreme and midlevel of ratings. Also, integration of SHAP with this model adds transparency and reliability factor to DistillBERT Model for getting linguistic pattern information which make system good for organizations and HR departments. Due to resource constraints, we have used DistillBERT in future more models can be explored like BERT. For expanding this work in future development and deployment of self-learning agents that works for improving model continuously over the time, and increasing relevance for both textual and contextual aspects. On the whole, combining better performance of Distill BERT with SHAP interpretable capability serves our solution as reliable and transparent for prediction of satisfaction score of employees

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